



Step 2A — Equilibrium Filter (Logical Formulation)

We define equilibrium as a combined condition across 3 dimensions:

1. Symmetry (Directional balance)

Measures whether buyers vs sellers dominate.

$$S = \frac{|n_{above} - n_{below}|}{n_{above} + n_{below}}$$

- Low S → balanced behavior → equilibrium-like
 - High S → directional pressure exists
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2. Central tendency distortion (shape stability)

Measures whether distribution is centered or drifting internally.

$$C = \frac{|mean - median|}{scale}$$

- Low C → symmetric / stable distribution
 - High C → skewed / structured distribution
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3. Structural persistence proxy (volatility-normalized dispersion)

Measures whether movement is meaningful or just noise.

$$V = \frac{scale}{iqr}$$

Interpretation:

- stable equilibrium → moderate, consistent ratio
 - chaotic noise → unstable extremes
 - structured regimes → deviation from baseline behavior
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Equilibrium Logic (Combined Condition)

Now combine them into a single equilibrium score:

$$E = w_1(1 - S) + w_2(1 - C) + w_3(\text{stability of } V)$$

But conceptually you don't even need weights yet.

Cleaner Decision Logic (no weights form)

Define equilibrium as:

Equilibrium = ALL of the following are true:

- Symmetry condition:

$$S \approx 0$$

- Center stability:

$$C \approx 0$$

- No structural deviation in scale:
 - V stays within its "normal band" (dataset-dependent baseline)
 - AND no persistence:
 - no directional dominance across the window
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FINAL EQUILIBRIUM FILTER (Decision Rule)

You classify a window as **equilibrium** → **NO TRADE** if:

- **Directional symmetry exists**
 - $|n_{above} - n_{below}|$ is small relative to window size

AND

- **No meaningful distribution skew**
 - $\text{mean} \approx \text{median}$ (scaled)

AND

- **No structural deviation in movement**
 - scale behaves like baseline (not expanding or compressing meaningfully)

AND

- **No persistence**
 - any imbalance present does not carry through time
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Intuition (what this filter is really doing)

You are removing windows where:

"price behaves like balanced random walk around a stable center"

In other words:

- no directional force
 - no structural deformation
 - no persistence
 - no regime energy
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Key Insight (important for your system)

This filter is NOT detecting “normal distribution”.

It is detecting:

absence of exploitable structure

That’s the correct framing for equilibrium in trading systems.